Module End Assignment 2_MySQL

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SELECT * FROM customer_churn;
SET SQL_SAFE_UPDATES = 0;

select
round(avg(WarehouseToHome),0) as MeanWarehouseToHome,
round(avg(HourSpendOnApp),0) as MeanHourSpendOnApp,
round(avg(OrderAmountHikeFromlastYear),0) as MeanOrderAmountHikeFromlastYear,
round(avg(DaySinceLastOrder),0) as MeanDaySinceLastOrder
from customer_churn;

UPDATE customer_churn
SET WarehouseToHome = COALESCE(WarehouseToHome, 16),
   HourSpendOnApp = COALESCE(HourSpendOnApp, 3),
   OrderAmountHikeFromlastYear = COALESCE(OrderAmountHikeFromlastYear, 16),
   DaySinceLastOrder = COALESCE(DaySinceLastOrder, 5);

SELECT Tenure
FROM customer_churn
WHERE Tenure IS NOT NULL
GROUP BY Tenure
ORDER BY COUNT(*) DESC
LIMIT 1;

SELECT CouponUsed
FROM customer_churn
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WHERE CouponUsed IS NOT NULL
GROUP BY CouponUsed
ORDER BY COUNT(*) DESC
LIMIT 1;

SELECT OrderCount
FROM customer_churn
WHERE OrderCount IS NOT NULL
GROUP BY OrderCount
ORDER BY COUNT(*) DESC
LIMIT 1;

UPDATE customer_churn
SET Tenure = COALESCE(Tenure, 1),
    CouponUsed = COALESCE(CouponUsed, 1),
    OrderCount = COALESCE(OrderCount, 2);

DELETE FROM customer_churn
WHERE WarehouseToHome > 100;

DELETE FROM customer_churn
WHERE CustomerID IN (
    SELECT CustomerID
    FROM (
        SELECT CustomerID
        FROM customer_churn
        WHERE WarehouseToHome > 100
    ) AS temp
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);

UPDATE customer_churn
SET PreferredLoginDevice = 'Mobile Phone'
WHERE PreferredLoginDevice = 'Phone';

UPDATE customer_churn
SET PreferedOrderCat = 'Mobile Phone'
WHERE PreferedOrderCat = 'Mobile';

UPDATE customer_churn
SET PreferredPaymentMode = 'Cash on Delivery'
WHERE PreferredPaymentMode = 'COD';

UPDATE customer_churn
SET PreferredPaymentMode = 'Credit Card'
WHERE PreferredPaymentMode = 'CC';

ALTER TABLE customer_churn
RENAME COLUMN PreferedOrderCat TO PreferredOrderCat;

ALTER TABLE customer_churn
RENAME COLUMN HourSpendOnApp TO HoursSpentOnApp;

ALTER TABLE customer_churn
ADD COLUMN ComplaintReceived VARCHAR(3);

UPDATE customer_churn
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SET ComplaintReceived =
   CASE
      WHEN Complain = 1 THEN 'Yes'
      ELSE 'No'
   END;

ALTER TABLE customer_churn
ADD COLUMN ChurnStatus VARCHAR(10);

UPDATE customer_churn
SET ChurnStatus =
   CASE
                  WHEN Churn= 1 THEN 'Churned'
      ELSE 'Active'
      END;

ALTER TABLE customer_churn
DROP COLUMN Churn,
DROP COLUMN Complain;

SELECT ChurnStatus, COUNT(*) AS CustomerCount
FROM customer_churn
GROUP BY ChurnStatus;

SELECT
   AVG(Tenure) AS AverageTenure,
   SUM(CashbackAmount) AS TotalCashbackAmount
FROM customer_churn
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WHERE ChurnStatus = 'Churned';


SELECT
   SUM(CASE WHEN ComplaintReceived = 'Yes' THEN 1 ELSE 0 END) * 100.0
   / COUNT(*) AS ComplaintPercentage
FROM customer_churn
WHERE ChurnStatus = 'Churned';


SELECT
   CityTier,
   COUNT(*) AS ChurnedCustomerCount
FROM customer_churn
WHERE ChurnStatus = 'Churned'
  AND PreferredOrderCat = 'Laptop & Accessory'
GROUP BY CityTier
ORDER BY ChurnedCustomerCount DESC
LIMIT 1;


SELECT
   PreferredPaymentMode,
   COUNT(*) AS CustomerCount
FROM customer_churn
WHERE ChurnStatus = 'Active'
GROUP BY PreferredPaymentMode
ORDER BY CustomerCount DESC
LIMIT 1;


SELECT
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    SUM(OrderAmountHikeFromlastYear) AS TotalOrderAmountHike
FROM customer_churn
WHERE MaritalStatus = 'Single'
  AND PreferredOrderCat = 'Mobile Phone';

  SELECT
    AVG(NumberOfDeviceRegistered) AS AverageDevicesRegistered
FROM customer_churn
WHERE PreferredPaymentMode = 'UPI';

SELECT
    CityTier,
    COUNT(*) AS CustomerCount
FROM customer_churn
GROUP BY CityTier
ORDER BY CustomerCount DESC
LIMIT 1;

SELECT
    Gender,
    SUM(CouponUsed) AS TotalCouponsUsed
FROM customer_churn
GROUP BY Gender
ORDER BY TotalCouponsUsed DESC
LIMIT 1;

SELECT
    PreferredOrderCat,
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    COUNT(*) AS CustomerCount,
    MAX(HoursSpentOnApp) AS MaxHoursSpentOnApp
FROM customer_churn
GROUP BY PreferredOrderCat;

SELECT
    SUM(OrderCount) AS TotalOrderCount
FROM customer_churn
WHERE PreferredPaymentMode = 'Credit Card'
  AND SatisfactionScore = (
      SELECT MAX(SatisfactionScore)
      FROM customer_churn
  );

  SELECT
    AVG(SatisfactionScore) AS AverageSatisfactionScore
FROM customer_churn
WHERE ComplaintReceived = 'Yes';

SELECT DISTINCT PreferredOrderCat
FROM customer_churn
WHERE CouponUsed > 5;

SELECT
    PreferredOrderCat,
    AVG(CashbackAmount) AS AverageCashback
FROM customer_churn
GROUP BY PreferredOrderCat
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ORDER BY AverageCashback DESC
LIMIT 3;

SELECT
    PreferredPaymentMode,
    AVG(Tenure) AS AverageTenure,
    SUM(OrderCount) AS TotalOrders
FROM customer_churn
GROUP BY PreferredPaymentMode
HAVING AVG(Tenure) = 10
   AND SUM(OrderCount) > 500;

SELECT
    CASE
        WHEN WarehouseToHome <= 5 THEN 'Very Close Distance'
        WHEN WarehouseToHome <= 10 THEN 'Close Distance'
        WHEN WarehouseToHome <= 15 THEN 'Moderate Distance'
        ELSE 'Far Distance'
    END AS DistanceCategory,
    ChurnStatus,
    COUNT(*) AS CustomerCount
FROM customer_churn
GROUP BY DistanceCategory, ChurnStatus
ORDER BY DistanceCategory, ChurnStatus;

SELECT *
FROM customer_churn
WHERE MaritalStatus = 'Married'
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  AND CityTier = 1
  AND OrderCount > (
      SELECT AVG(OrderCount)
      FROM customer_churn
  );



CREATE TABLE customer_returns (
    ReturnID INT PRIMARY KEY,
    CustomerID INT,
    ReturnDate DATE,
    RefundAmount DECIMAL(10,2)
);

INSERT INTO customer_returns
    (ReturnID, CustomerID, ReturnDate, RefundAmount)
VALUES
    (1001, 50022, '2023-01-01', 2130),
    (1002, 50316, '2023-01-23', 2000),
    (1003, 51099, '2023-02-14', 2290),
    (1004, 52321, '2023-03-08', 2510),
    (1005, 52928, '2023-03-20', 3000),
    (1006, 53749, '2023-04-17', 1740),
    (1007, 54206, '2023-04-21', 3250),
    (1008, 54838, '2023-04-30', 1990);

SELECT *
FROM customer_returns AS r
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INNER JOIN customer_churn AS c
    ON r.CustomerID = c.CustomerID
WHERE c.ChurnStatus = 'Churned'
  AND c.ComplaintReceived = 'Yes';
```